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Excel Syllabus

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Meaning

- worksheet, rows, columns, cell
- why we need excel, types of data

Important features of excel.

- 1 Data cleaning
- 2 formulas and function
- 3 Data visualization
- 4 Pivot tables
- 5 Sorting and filtering
- 6 Data validation
- 7 Pivot tabel
- 8 Sorting and filtering
- 9 Data validation

Meaning

Microsoft excel is a spreadsheet software used to store, organize, calculate, analyze and visualize data. It is a beginner friendly

workbook: a workbook is the entire excel file with (.xlsx) extension

worksheet: a worksheet is an individual sheet inside the workbook

row: run horizontally and are numbered: 1, 2, 3, 4...

columns: runs vertically and are named A, B, C, D
cell: The intersection of row and column is called a cell

Why we need excel

- * Stores large amount of data
- * Clean data, perform calculation
- * analyze data, create charts
- * finds patterns and trends
- * No need of coding

Types of data in Excel

- 1 Text data: contains letters, word, characters
- 2 Number data: no. are used for calculation
- 3 Date data: date represents a specific day, month, year
- 4 Time data: Represents a specific time
- 5 Currency data: Represents money value
- 6 Boolean data: Logical data with two possible values (T/F, 0/1)

Important excel features

- 1 Data cleaning
 - * This finds and corrects problems in raw data
 - * removing duplicate records

- * corrects missing values, spellings, extra spaces
- * Incorrect data format

2 formulas and functions

A formula is an expression we create to perform a calculation

SUM, AVERAGE, COUNT, COUNTA, MAX, MIN

3 Data Visualization

Representing data ~~visu~~ visually using charts and graphs like bargraph, piechart

4 Pivot table

Allows to summarize a large amt of data quickly very important

5 Sorting and filtering

- * arranging data smallest to largest, largest to smallest

- * filtering means showing only the data we want while hiding the rest temporarily

6 Data validation

This allows and controls what a users are allowed to enter in a cell

Advantage

- Easy to learn
- Powerful calculation
- Data visualization & Data analysis
- widely used & flexible

Disadvantage

- can become difficult with huge datasets
- Manual errors
- Collaboration limitations
- Security concerns
- Repetitive work

functions

set of instruction / set of code is already predefined is called function

there are 3 main components of functions

- 1 function name
- 2 number of inputs
- 3 return values

Excel functions

1. **upper()** : This function converts all letters in a text into capital letters

Syntax : = UPPER (text)

Ex : = UPPER (A2), UPPER ("Sahana")

o/p : SAHANA

2. **lower()** : converts all letters in a text into small letters

Syntax : = LOWER (text)

Ex : = LOWER (A2), = LOWER ("SAHANA")

o/p : sahana

3. **Proper()** : This converts the first letter of each word into capital letter and remaining letters into small letters

Syntax : = PROPER (text)

Ex : PROPER (A2), = PROPER ("SAHANA")

o/p : Sahana.

- 4 **LEN()**: This counts the no. of characters in a text and also spaces are counted as a char

Syntax: =LEN (text)

Ex: =LEN (A2), =LEN ("Sahana")

o/p: 6

- 5 **MID()**: This function extracts a specific no of characters from the middle of a text, starting from a position we specify

Syntax: =MID (text, start - num, num - chars)

Ex: =MID (A2, 2, 3), =MID ("Sahana", 2, 3)

OP: aha

- 6 **LEFT()**: This extracts a specified no. of characters from the left side of the text

Syntax: =LEFT (text, num - char)

Ex: =LEFT (A2, 3), =LEFT ("Sahana", 3)

o/p: sah

- 7 RIGHT(): extracts a specified no. of characters from the right side of a text

Syntax: =RIGHT (text, num_chars)

Ex: =RIGHT (A2, 3), =RIGHT ("Sahana", 3)

O/p: ana

- 8 Dividing the text into 1st half and 2nd half.

- used when we need to split a text into 2 parts

for even numbers: ANUSHA (6 chars)

1st half Syntax: =LEFT (text, LEN (text)/2)

Ex: =LEFT (A2, LEN (A2)/2)

O/p: ANU

2nd half Syntax: =RIGHT (text, LEN (text)/2)

Ex: =RIGHT (A2, LEN (A2)/2)

O/p: SHA

for odd numbers: SAHANA (6 chars)

when the text has an odd no. of chars it cannot be divided into equal halves. So, ROUNDUP() function should be

used in any one of the half

1st half Syntax: =LEFT (text, ROUNDUP (LEN (text) / 2, 0))
Ex: =LEFT (A2, ROUNDUP (LEN (A2) / 2, 0))
O/p: SAHA

2nd half Syntax: =RIGHT (text, LEN (text) / 2)
Ex: =RIGHT (A2, LEN (A2) / 2)
O/p: ANA